EXPERIMENT 7

Aim: Write a Program to implement Sliding Window Protocol for Selective Repeat ARQ.

SOLUTION:

Algorithm: Selective Repeat ARQ

Step 1: Start.

Step 2: Read the total number of frames (**n**).

Step 3: Read the sender window size (**w**).

Step 4: Send all frames within the current sender window.

Step 5: Receive an ACK for each frame.

Step 6: If a frame is acknowledged, mark it as acknowledged.

Step 7: If a frame is lost or not acknowledged, retransmit **only that frame**.

Step 8: Slide the sender window forward only when the first frame in the current window has been acknowledged.

Step 9: Repeat until all frames are transmitted successfully.

Step 10: Stop.

Source Code:

```
#include <stdio.h>
#include <conio.h>

int main()
{
    int n, w;
    int ack[100] = {0};
    int i = 0, j, status;

   clrscr();

    /* Read total number of frames */
    printf("Enter the total number of frames: ");
    scanf("%d", &n);

    /* Read sender window size */
    printf("Enter the sender window size: ");
    scanf("%d", &w);

    while (i < n)
    {
        printf("\nSender Window: ");

        /* Display current sender window */
        for (j = i; j < i + w && j < n; j++)
        {
```

```
            printf("%d ", j);
        }

        printf("\n");

        /* Send all unacknowledged frames */
        for (j = i; j < i + w && j < n; j++)
        {
            if (ack[j] == 0)
            {
                printf("Frame %d sent.\n", j);
            }
        }

        /* Receive ACK for each frame */
        for (j = i; j < i + w && j < n; j++)
        {
            if (ack[j] == 0)
            {
                printf("Enter ACK for Frame %d (1 = ACK, 0 = Lost): ", j);
                scanf("%d", &status);

                if (status == 1)
                {
                    ack[j] = 1;
                    printf("Frame %d acknowledged.\n", j);
                }
                else
                {
                    printf("Frame %d lost.\n", j);
                }
            }
        }

        /* Retransmit only lost frames */
        for (j = i; j < i + w && j < n; j++)
        {
            while (ack[j] == 0)
            {
                printf("\nTimeout for Frame %d.\n", j);
                printf("Retransmitting only Frame %d...\n", j);

                printf("Enter ACK for retransmitted Frame %d (1 = ACK, 0 =
Lost): ",
                       j);
                scanf("%d", &status);

                if (status == 1)
                {
                    ack[j] = 1;
                    printf("Frame %d acknowledged.\n", j);
                }
                else
                {
                    printf("Frame %d lost again.\n", j);
```

```c
                }
            }
        }

        /* Slide the window */
        while (i < n && ack[i] == 1)
        {
            i++;
        }
    }

    printf("\nAll frames are transmitted successfully.");

    getch();
    return 0;
}
```

Output: